FOR A FIBRED CATEGORY OVER A SITE THE LOCALISED MORPHISM OF TOPOI ADMITS A SECTION WHOSE PULLBACK IS THE PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the construction of the strongly cartesian functor v, its continuity with right adjoint p, the identification of pi_{U,*} with the pullback along v, and the cocontinuity argument producing the section sigma and the identities sigma^{-1} = pi_{U,*} and pi_U sigma = id.

Editorial note. The label lemma-fibred-category-with-object names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the standing situation of a fibred category; its four cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Situation of a fibred category over a site and the induced morphism of topoi pi_U. Same chapter, the standing situation of a fibred category p : C -> D over a site and the localised morphism pi_U : Sh(C/U) -> Sh(D/p(U)). Supplies the morphism of topoi for which a section is constructed.
- Strongly cartesian morphisms and continuity of the associated functor. Sites, the localisation of a morphism of topoi and the discussion of the cocontinuous-adjoint situation for a continuous right adjoint functor. Supplies that the functor V |-> V x_{p(U)} U is continuous with right adjoint p, and hence the description of pi_{U,*}.

Source reference. Original lemma 00XY: sources/stacks-chow/tags.txt, source lines 1236–1236. Prerequisite pointer invoked at proof line(s) 9752; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03EG: sources/stacks-chow/tags.txt, source lines 4455–4455. Prerequisite pointer invoked at proof line(s) 9736; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original section 00XW: sources/stacks-chow/tags.txt, source lines 1234–1234. Prerequisite pointer invoked at proof line(s) 9748; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08P8: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9701–9701. Prerequisite pointer invoked at proof line(s) 9721; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assumptions and notation as in Situation 08P8. For $U \in \mathrm{Ob}(\mathcal{C})$ consider the induced morphism of topoi*

$$\pi_U : Sh(\mathcal{C}/U) \longrightarrow Sh(\mathcal{D}/p(U))$$

Then there exists a morphism of topoi

$$\sigma : Sh(\mathcal{D}/p(U)) \to Sh(\mathcal{C}/U)$$

such that $\pi_U \circ \sigma = id$ and $\sigma^{-1} = \pi_{U,*}$.

Proof. Observe that π_U is the restriction of π to the localizations, see Sites, Lemma 03EG. For an object $V \to p(U)$ of $\mathcal{D}/p(U)$ denote $V \times_{p(U)} U \to U$ the strongly cartesian morphism of $\mathcal{C}$ over $\mathcal{D}$ which exists as p is a fibred category. The functor

$$v : \mathcal{D}/p(U) \to \mathcal{C}/U, \quad V/p(U) \mapsto V \times_{p(U)} U/U$$

is continuous by the definition of the topology on $\mathcal{C}$. Moreover, it is a right adjoint to p by the definition of strongly cartesian morphisms. Hence we are in the situation discussed in Sites, Section 00XW and we see that the sheaf $\pi_{U,*}\mathcal{F}$ is equal to $V \mapsto \mathcal{F}(V \times_{p(U)} U)$ (see especially Sites, Lemma 00XY).

But here we have more. Namely, the functor v is also cocontinuous (as all morphisms in coverings of $\mathcal{C}$ are strongly cartesian). Hence v defines a morphism σ as indicated in the lemma. The equality $\sigma^{-1} = \pi_{U,*}$ is immediate from the definition. Since $\pi_U^{-1}\mathcal{G}$ is given by the rule $U'/U \mapsto \mathcal{G}(p(U')/p(U))$ it follows that $\sigma^{-1} \circ \pi_U^{-1} = \mathrm{id}$ which proves the equality $\pi_U \circ \sigma = \mathrm{id}$. □